

Subject: NBM-25-0304 - Decision on Manuscript

From: NMR in Biomedicine <onbehalf@manuscriptcentral.com>

Date: 2025-04-11, 3:24 a.m.

To: Alexander Weber <aweber@bcchr.ca>

CC: "annaleezhu@gmail.com" <annaleezhu@gmail.com>

11-Apr-2025

Dear Dr. Weber,

Manuscript # NBM-25-0304 entitled "Assessing Regional Cerebral Oxygen Consumption (CMRO2) in Preterm Neonates: A Quantitative MRI Cohort Study with Exploratory Analysis of Respiratory Support" which you submitted to NMR in Biomedicine, has been reviewed. The comments of the reviewers are included at the bottom of this letter. The reviewers have recommended major revisions to your manuscript. Therefore, I invite you to respond to the reviewers' comments and revise your manuscript. Please note that resubmitting your manuscript does not guarantee acceptance.

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I look forward to receiving your revision.

Sincerely, Prof. John Griffiths
Editor, NMR in Biomedicine
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Comments to Author:

Reviewer: 1

Comments to the Author

This manuscript reports CMR02 assessments in preterm neonates. The findings are clinically meaningful. However, CMR02 or OEF algorithm from QSM is not clearly described. The authors may consider the use of OEF mapping from QSM data as established in literature.

Specific comments

Results

"Mean whole-brain CSv02, CSa02, Hct and oxygen extraction fraction (OEF) values were": how were these values measured? I cannot find in the manuscript any description on CSa02 and OEF measurements.

"D is simply B multiplied by a value determined by CSa02, CSv02, and Hct": how is the value determined – equation?

Discussion

"measure the CSv02 by averaging over all internal veins." "Our CSv02 processing pipeline filtered out χ values below 0.15 ppm in order to obtain realistic values." These statements should be provided in methods to describe processing algorithm.

References

Cite recent consensus paper on QSM, PMID: 38247051

OEF mapping from QSM data: Kudo et al, PMID: 26661168; Cho et al, PMID: 29516537.

Reviewer: 2

Comments to the Author

This study reported cerebral metabolic rate of oxygen (CMR02) measurements in preterm neonates who underwent respiratory therapy. The authors quantified CMR02 (and perfusion, venous oxygenation) values in these neonates and investigated their relationship to ventilation therapy duration and category. Given how challenging it is to measure CMR02 in preterm neonates, this study is highly novel and provides some unique data to the literature. The manuscript was clearly written. The study is generally sound. My main comments are on the details of the methods, in particular the description of the SV02 estimation and related results.

1. Methods: It would be useful to clarify whether the MRI scan was research-dedicated scan or clinical scan with research sequences added (clinical add-on).

2. Methods: for pCASL imaging parameter, it's unclear why 50 slices were acquired. 50 slices * 4mm = 200 mm. That will cover down to the shoulder of the neonate. Then where is the labeling

plane. Please clarify. Also, "pulse label delay" should be "post-labeling delay".

3. Methods: SWI had a $1 \times 1 \times 2 \text{ mm}^3$ voxel size. At this voxel size in neonates, if the venous ROI contains any tissue partial voluming. It would result in an over-estimation in venous oxygenation, or under-estimation in OEF and CMR02.

4. Methods: The data processing for ASL, SV02, and CMR0 estimation were too brief. I could not find the supplemental file actually. In any way, these processing procedures should be included in the main text, as they are critical for the evaluation of the validity of the results.

5. Methods: Related to the above comment, the authors should clarify whether the SV02 result is a voxel-by-voxel map or a region of interest result. If ROI only, then how can one obtain a CMR02 map. This is a bit puzzling.

6. Results: In Figure 4, CMR02 map is the same as the perfusion map. A single, global SV02 value was used. Then how is the QSM based method better than the global SV02 methods? Note that the results below talked about regional CMR02. This is also confusing.

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